

Ahmed Toba Mahmoud

Software Engineer — Backend Developer

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Professional Summary

Software Engineer with strong foundations in data structures, algorithms, OOP, and system design, specializing in scalable, secure backend systems using Node.js, Express.js, RESTful APIs, and GraphQL. Experienced in modular service-layered architecture, automated testing, and production security hardening, with a focus on building reliable, maintainable systems.

Education

B.Sc. in Communications and Electronics Engineering | Beni Suef University | 2026 | Grade: Very Good

Graduation Project | Forging a Secure and Resilient Digital Backbone for a New Era of Academia

Grade: Excellent

[GitHub](#) | [Live Demo](#)

- Led a multidisciplinary engineering team in planning, architecting, and delivering a secure Smart University ecosystem spanning Backend Development, Cloud & DevOps, AI, IoT, Enterprise Networking, and Cybersecurity.
- Designed and implemented scalable backend services using Node.js and Express.js, RESTful APIs, real-time communication using Socket.IO, authentication and authorization mechanisms, and cloud-integrated system architecture.
- Developed a cloud-connected biometric attendance system using ESP32, Azure IoT Hub, secure device communication, offline synchronization, and real-time attendance processing. [GitHub](#)
- Engineered a multi-layer AI academic assistant featuring general academic Q&A, role-based live data retrieval, and a RAG-powered document intelligence system, with token budget optimization and defenses against prompt injection.

Professional Experience

Backend Developer Intern | Hi Deep AI | Beni Suef, Egypt

Sep 2025 – Dec 2025

- Collaborated directly with clients across the full Software Development Lifecycle (SDLC), contributing to business analysis, system analysis, solution architecture, implementation, testing, and project delivery.
- Designed, developed and enhanced scalable backend services, RESTful APIs, MongoDB data models, file upload functionality, and core application features within an Agile development team.
- Wrote unit tests, participated in code reviews, and resolved technical issues to improve software quality.

Projects

Natours — Full-Stack Tour Booking Platform [GitHub](#) | *Node.js · Express.js · MongoDB · JWT · Stripe · Pug*

- Built full-stack tour booking platform featuring secure RESTful APIs, server-side rendering (SSR), and responsive.
- Implemented JWT authentication, Stripe, rate limiting, XSS/NoSQL protection, and other security best practices.

E-Commerce Platform — Full-Stack Shopping Platform [GitHub](#) | *Angular · Node.js · Express.js · MongoDB*

- Developed a role-based full-stack e-commerce platform with product, cart, order, and authentication modules.
- Implemented secure backend APIs and integrated frontend workflows for a complete shopping experience.

GraphQL Blog API — Production-Hardened GraphQL API | [GitHub](#) | *Node.js · GraphQL · MongoDB · Mocha*

- Architected a GraphQL API using a modular, service-layered design, with 10+ production security layers.
- Built a comprehensive automated test suite (13+ files) covering unit, integration, and E2E testing using Mocha, Chai, Sinon, and Supertest with isolated test database and automated teardown hooks.

Training & Certifications

- NodeJS - The Complete Guide (MVC, REST APIs, GraphQL, Deno)** – Udemy (Max) 2026
- MEAN Stack Web Development (120 Hours)** – National Telecommunication Institute (NTI) 2025
- Node.js, Express, MongoDB & More: The Complete Bootcamp** – Udemy (Jonas Schmedtmann) 2025

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript, C++, C, SQL
- **Backend Development:** Node.js, Express.js, RESTful APIs, GraphQL, API Security, Socket.IO.
- **Frontend Development:** HTML5, CSS3, Bootstrap, Angular (Basic), React (Basic), EJS, Pug
- **Databases:** MongoDB, MySQL, Mongoose, Sequelize
- **Testing:** Mocha, Chai, Sinon, Supertest
- **Security & Production:** Helmet, Rate Limiting, HPP, XSS/NoSQL Sanitization, Winston Logging, Nodemailer
- **Cloud & DevOps:** Microsoft Azure (Basic), Azure IoT Hub, Docker, AWS (Basic), Git, GitHub
- **AI Engineering:** RAG, Prompt Engineering, Prompt Injection Mitigation, Role-Based AI Systems, Token Budget Optimization, LLM Integration
- **IoT Development:** ESP32 Firmware Development, MQTT, Biometric Integration, IoT Device Communication
- **Core Fundamentals:** OOP, Data Structures & Algorithms, SOLID, Software Design, Modular Architecture
- **Developer Tools:** Postman, Linux

Leadership & Volunteering

Co-Founder & Business Developer | Maymay Start-up Jan 2025 – Jan 2026

- Co-led business strategy, product development, and partnership initiatives to support startup growth.

Rally BSU | Beni Suef University

HR Manager Aug 2025 – Aug 2026

Led recruitment, team development, conflict resolution, and HR operations.

Marketing Vice Feb 2025 – Aug 2025

Directed marketing initiatives, campaign planning, and cross-functional team coordination.

Head of HR | GDSC EELU Feb 2024 – Aug 2024

- Managed recruitment, onboarding, mentoring, and student progress throughout Google Developer Student Clubs learning programs.

Languages

- Arabic:** Native
- English:** Professional Working Proficiency